

Software Requirements Specification

for

Financial Document Intelligence System

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1. Introduction

1.1 System Description

Financial Document Intelligence System is a platform designed to help users quickly understand large financial documents — such as annual reports, financial statements and bank statements by generating summaries, retrieving important information, and highlighting key insights.

1.2 System Scope

The Financial Document Intelligence System allows:

- To upload financial documents in digital format.
- To search and retrieve important information from uploaded documents.
- Generation of summaries of financial documents
- Users to ask questions related to uploaded documents
- Highlight key insights and relevant sections from the documents
- Reduction of manual effort in reading and analyzing large financial documents.

1.3 Intended Audience

The intended audience of this document include:

- Project Team: Developers, Testers, Project Managers
- Stake Holders: Mentors, Evaluators, Organization Management

1.4 Module Scope

This implementation focuses on the Financial Document Analysis and Retrieval module of the system. The module is responsible for processing uploaded financial documents, retrieving relevant information based on user queries, generating summaries, and highlighting key insights from the documents. The implementation mainly focuses on improving document search and analysis workflows for users working with financial information.

1.5 Assumptions

- The uploaded documents are assumed to primarily contain text-based content in English.
- The uploaded documents are assumed to be in pdf or spreadsheet (excel) format.
- Users are assumed to have basic knowledge of financial documents and related terminology.
- The system is intended for informational and research purposes only and does not provide financial advice.
- Authentication and advanced user management features are considered outside the scope of this implementation.
- The number of uploaded documents remain within manageable limit for prototype implementation
- The responses generate by the system may require human verification for critical decision making.

1.6 Definitions and Acronyms

- **PDF** – Portable Document Format. The primary file format for financial documents in this system.
- **XLSX** – Spreadsheet Format (Microsoft Excel or other compatible spreadsheet applications)
- **NLP** – Natural Language Processing. NLP libraries are used to retrieve information from the uploaded documents.
- **Supported Documents** –
 - i) Annual reports
 - ii) Quarterly Reports
 - iii) Financial Statements
 - Balance Sheets
 - Income Statements
 - Cash Flow Statements
 - iv) SEC Filings
 - v) Audit Reports
- **Supported File Formats** –
 - i) PDF
 - ii) XLSX

2. Functional Requirements

2.1 Document Upload

Users must upload financial documents in digital format for analysis and processing. Uploaded documents will be used by the system for information retrieval, summary generation and respond to user queries.

Acceptance Criteria:

1. Users must be able to upload financial documents through the system interface.
2. The system must accept supported document formats such as PDF files.
3. Invalid or unsupported file formats must be rejected with an appropriate message.
4. Uploaded documents must be stored successfully for further processing and analysis.
5. The system must notify the user whether the document upload was successful or unsuccessful.

2.2 Document Processing

The system must process uploaded financial documents by extracting and preparing the document content for analysis and information retrieval. The processed content will be used for generating summaries, retrieving relevant information, and responding to user queries.

Acceptance Criteria:

1. The system must extract text content from uploaded financial documents.
2. The system must process the extracted content for further analysis and retrieval operations.
3. Documents with unreadable or invalid content must be identified and handled appropriately.
4. Successfully processed documents must be made available for querying and summarization.
5. The system must notify the user if document processing fails.

2.3 Information Retrieval

The system must allow users to retrieve relevant information from uploaded financial documents based on user queries. The system should identify and present the most relevant sections of the documents related to the requested information.

Acceptance Criteria:

1. Users must be able to search for information using queries
2. The system must retrieve relevant content from the uploaded financial documents based on user query.
3. The retrieved information must be displayed in a clear and understandable format.
4. Handle situation where relevant information cannot be found
5. Retrieved result must correspond to the uploaded documents available in the system.

2.4 Query-based Response Generation

The system must generate responses to user queries based on the information retrieved from uploaded financial documents. The generated responses should provide relevant and understandable answers related to the user's request.

Acceptance Criteria:

1. Users must be able to submit queries related to uploaded financial documents using a chatbot interface.
2. The system must generate responses based on the retrieved document content.
3. The system must present responses in a clear and understandable format relevant to user query.
4. The system must notify the user appropriately if a valid response cannot be generated.
5. The system shall not retain or provide direct access to previously uploaded documents after session completion; however, summarized insights and key discussion points generated during prior interactions may remain accessible within the conversation context.

2.5 Document Summarization

The system must generate concise summaries of uploaded financial documents to help users quickly understand the main information and key points contained in the documents.

Acceptance Criteria:

1. Users must be able to request summaries for uploaded financial documents.
2. The system must generate summaries based on the content of the uploaded documents.
3. The generated summaries must present the important information in a concise and understandable manner.
4. The system must handle cases where document summarization cannot be completed successfully.
5. The generated summaries must correspond to the content of the uploaded financial documents.

2.6 Insight Highlighting

The system must identify and highlight important insights and relevant sections from uploaded financial documents to help users quickly locate significant information.

Acceptance Criteria:

1. The system must identify key information and important sections from uploaded financial documents.
2. The system must present highlighted insights in a clear and understandable format.
3. Users must be able to view the highlighted sections within the retrieved results or summaries.
4. The system must handle cases where significant insights cannot be identified.

3. Non-Functional Requirements

3.1 Performance

The system should process uploaded financial documents and generate responses within a reasonable amount of time to ensure a smooth user experience.

3.2 Logging and Error Handling

The system should maintain logs for important operations and handle errors appropriately to prevent system failures during document upload, processing, and query handling.

3.3 Input Validation

The system should validate all uploaded documents and user queries before processing. Invalid or unsupported inputs should be identified and handled with appropriate messages.

3.4 Reliability

The system should consistently retrieve relevant information and generate meaningful responses based on uploaded financial documents.

3.5 Usability

The system should provide a simple and user-friendly interface that allows users to upload documents, search information, and view summaries easily.

3.6 Security

The system should ensure safe handling of uploaded documents and user queries. Sensitive information should not be exposed through system outputs or logs.

4. Other Requirements

4.1 Operational Environment

The implementation of this system is carried out using Python as the primary programming language. The document processing and analysis functionalities are supported using AI and NLP libraries. The system utilizes a vector database for storing and retrieving processed document information. The application is intended to run in a standard development environment with internet connectivity for AI-based processing services.

5. Future Scope

The following enhancements and features may be considered for future development of the system:

- Integration with real-time financial data and market information sources.
- Advanced analytics and financial trend analysis features.
- Enhanced insight generation and visualization dashboards.
- Multi-user support with authentication and access control features.
- Integration with external financial research and reporting platforms.
- Improved document processing capabilities for scanned or image-based documents.
- Support for additional document formats and multilingual documents.